CCIT Report #505
October 2004

Optimal Bus Sizing in Migration of Processor Design

Shmuel Wimer, Shay Michaely, Konstantin Moiseev, Avinoam Kolodny

Abstract

The effect of wire delay on circuit timing typically increases when an existing layout is migrated to a new generation of process technology, because wire resistance and cross capacitances do not scale well. Hence, careful sizing and spacing of wires is an important task in migration of a processor to next generation technology. In this paper, timing optimization of signal buses is performed by resizing and spacing individual bus wires, while the area of the whole bus structure is regarded as a fixed constraint. Four different objective functions are defined and their usefulness is discussed in the context of the layout migration process. The paper presents solutions for the respective optimization problems and analyzes their properties. In an optimally-tuned bus layout, after optimizing the most critical signal delay, all signal delays (or slacks) are equal. The optimal solution of the MinMax problem is always bounded by the solution of the corresponding sum-of-delays problem. An iterative algorithm to find the optimally-tuned bus layout is presented. Examples of solutions are shown, and design implications are derived and discussed.

Introduction

Interconnect delays have become dominant in CMOS VLSI digital systems as a result of technology scaling [1] [2]. In recent generations, wire resistance and cross-capacitance between adjacent wires have become increasingly important in their effect on signal delay. For a given metal layer, wire resistance and cross-capacitance depend on wire width and inter-wire spacing, respectively. Allocation of wire widths and spaces for bus structures under a total area constraint is an important problem in process migration of existing mask layouts (also known as "process shifting"), which often produces excessive wire delays in the new layout. The strategy of allocating widths and spaces to maximize performance in bus structures was proposed in [3] without formal analysis and solution. The nature of this problem allows tradeoff between the resistance of a wire and its coupling capacitances to adjacent wires, by increasing wire width while reducing spaces, or vice versa. Wire resistance affects only the delay of the signal carried by the wire, while coupling capacitances affect the delays of both the wire and its neighbors. For multiple nets, the optimal solution involves simultaneous tradeoffs among all wires sharing a given common area.

The wire sizing problem has been addressed in [4] and [5] for a single wire and for a single-net interconnect tree. Simultaneous wire sizing and driver sizing has been presented in [6, 7]. The problem of sizing and spacing multiple nets with consideration of coupling capacitance in global interconnect has been addressed in [8], considering general tree structures for nets with fixed terminals, without a total area constraint. The authors modeled coupling between nets by converting cross-capacitance into an effective fringe

capacitance, which resulted in a decoupled delay model for each net. The routing tree for each net was sized independently, using an algorithm based on dynamic programming [9]. Coupling capacitance has been considered more explicitly in the context of physical design algorithms for minimizing crosstalk noise [4,10,11] or dynamic power [12]. The authors of [13] derived layout guidelines and presented a simultaneous multiple-net spacing algorithm for area minimization in general layouts under a noise-constraint.

This paper addresses the problem of simultaneously assigning widths and spaces to n parallel wires, representing a bus or several interleaved busses, as illustrated in Figure 1. Such geometry is commonly used in practice, and its simplicity enables straightforward mathematical analysis. With given drivers, load capacitances and timing requirements for the individual signals, wire widths and spaces are allocated to maximize circuit speed. Note that driver strengths, load capacitances and required arrival times are not necessarily equal. The total sum of widths and spaces is a given constraint, representing the total width available for the bus structure in the layout.

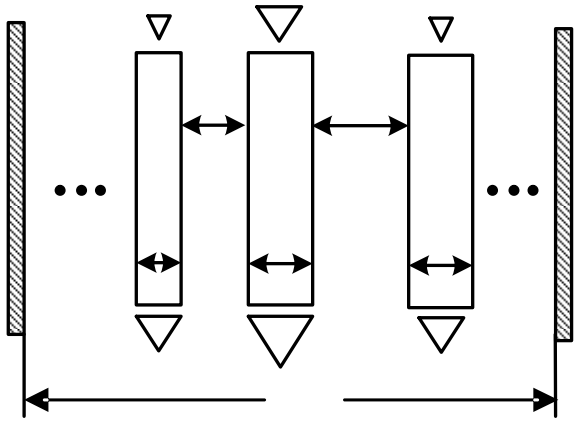


Figure 1. Structure of the bus: n parallel signal wires share a fixed total width A between two shield wires.

Problem Formulation

Consider a bus of n signal nets $\sigma_1,\ldots,\sigma_n$ between two side-walls (wires at fixed locations, connected to V_{cc} or V_{ss}) as shown in Figure 1. S_{i-1} and S_i, respectively, denote spaces to the right and left neighbors of wire W_i. The length of each wire is L. The sum of wire widths and spaces between the left and right side walls is given in the following constraint, which represents the total width A of the available area for laying out the signal bus.

$$(1)\ g\left(\overline{W},\overline{S}\right)=\sum_{j=1}^{n} W_i+\sum_{j=0}^{n} S_i = A\ .$$

Another set of constraints on wire sizing is geometrical design rules, which are imposed by the manufacturing technology. In modern processes of 90 nanometers and below, the width and the space of wires are bounded in some range as follows:

(2) $S' \le S_i \le S'', 0 \le i \le n$, and

(3) $W' \le W_i \le W'', 1 \le i \le n$.

Delay model

Signal delays are expressed by an Elmore model using first order approximation for capacitances. The delay of signal σ_i can be calculated from the π-model equivalent circuit shown in Figure 2, where R_{d_i} is the effective output resistance of the driver, R_{w_i} is the wire resistance, C_{w_i} is the wire area and fringe capacitance, $C_{c_{i-1}}$ and C_{c_i} are coupling capacitances to the right and left neighboring signals, and C_{l_i} is the capacitive load presented by the receiver's input. Using technology parameters these can be expressed as $R_{R_i} = R_s L/W_i$, $C_{w_i} = C_a L W_i + C_f L$ and $C_{c_i} = k_c L/S_i$, where C_a is area capacitance coefficient, C_f is fringe capacitance coefficient, k_c is a line-to-line coupling coefficient, and R_s is the metal sheet resistance.

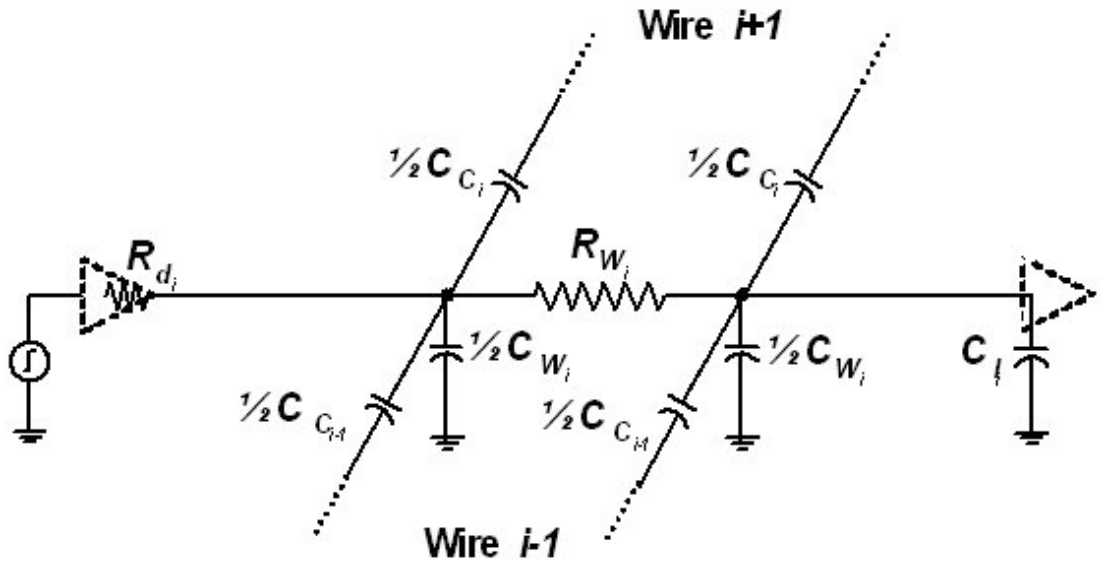


Figure 2. Equivalent circuit for calculating the i[th] signal delay

Under Elmore delay model, the delay Δ_i of signal σ_i from driver's input to receiver's input is given as follows:

$$\Delta_i = R_{d_i} C_a L W_i + \left(D_i + R_{d_i} C_f L + R_{d_i} C_{l_i} + \frac{1}{2} C_a R_s L^2 \right) + \left(\frac{1}{2} R_s C_f L^2 + R_s L C_{l_i} \right) \frac{1}{W_i} + R_{d_i} k_c L \left(\frac{1}{S_{i-1}} + \frac{1}{S_i} \right) + \frac{1}{2} R_s k_c L^2 \frac{1}{W_i} \left(\frac{1}{S_{i-1}} + \frac{1}{S_i} \right) \quad (4)$$

The coefficients of wire width and spaces in (1) will be marked as a_i, b_i, c_i, d_i, e. The delay expression can be rearranged as

$$\Delta_i \left(\overline{W}, \overline{S} \right) = a_i W_i + b_i + \frac{c_i}{W_i} + \left(d_i + \frac{e}{W_i} \right) \times \left(\frac{1}{S_{i-1}} + \frac{1}{S_i} \right). \quad (5)$$

Note that in (5) the coefficient e is not indexed since it encapsulates only technology parameters, which are common to all delays. The other coefficients are indexed since they include parameters related to the signal's driver and receiver.

Despite its simplicity, this Elmore-based modeling approach is widely used in interconnect optimization. In [14], good accuracy versus circuit simulation has been obtained by applying a parameter fitting procedure to a similar wire delay model, in which the cross-capacitances were replaced by a fringing-field term. Note that the cross coupling capacitances between wires are not multiplied by a "Miller factor" [15] in the model equations above. Hence, nominal delays are considered, without delay uncertainty induced by crosstalk. In a worst-case design approach, the cross-capacitance terms are typically multiplied by a Miller factor of 2. For such a case, inter-wire tradeoffs become even more pronounced in optimizing the bus layout.

Sensitivity of signal delay to wire width and spaces

Consider a single wire placed between two side-walls. The delay of the wire is given by (4), with $i = 1$. Partial derivatives with respect to W and S_0 are as follows (note that the delay function is symmetrical in S variables):

$$(6)\ \frac{\partial \Delta}{\partial W} = R_d C_a L - \frac{1}{W^2}\left(\frac{1}{2} R_s C_f L^2 + R_s C_l L + \frac{1}{2} R_s k_c L^2 \left(\frac{1}{S_0} + \frac{1}{S_1}\right)\right)$$

$$(7)\ \frac{\partial \Delta}{\partial S_0} = -\frac{1}{S_0^{\,2}}\left(R_d k_c L + \frac{1}{2} R_s k_c L^2 \frac{1}{W}\right)$$

For each specific value of S the sensitivity to W is zero at a certain point $(W(\Delta_{\min}), S)$. This point is the minimum delay point for the given value of S. Sensitivity to S decreases monotonically with increasing of S and W.

In layout migration, wire width and spaces to neighbors cannot change independently. The additional constraint applied to W and S is:

(8) $W + S_0 + S_1 = A$, and therefore,

(9) $$\frac{\partial \Delta}{\partial W} = -\frac{\partial \Delta}{\partial S_0}.$$

The sensitivities to both S and W are thus identical. An example is shown in Figure 3 using 90 nanometer technology parameters for different driver resistances – 100Ω, 500Ω and 1000Ω, driving a wire of 1000 microns length, with load capacitance of 50 fF and the distance between walls is 1.5 microns. Sensitivity to both W and S was calculated for values of W from 0 to 1.5 μm. At the minimum delay point, sensitivity to wire width and spaces is zero, because the effect of any change in wire resistance balances out with the respective change in capacitances. A similar balance is obtained also when a bus with multiple wires is optimized, as will be discussed below. For wide buses, where inter-wire separation is large, the optimal width for each wire depends mostly on values of driver resistance and load capacitance of the wire, according to (8). This may be used as a first approximation for assigning initial values to wire widths in bus optimization.

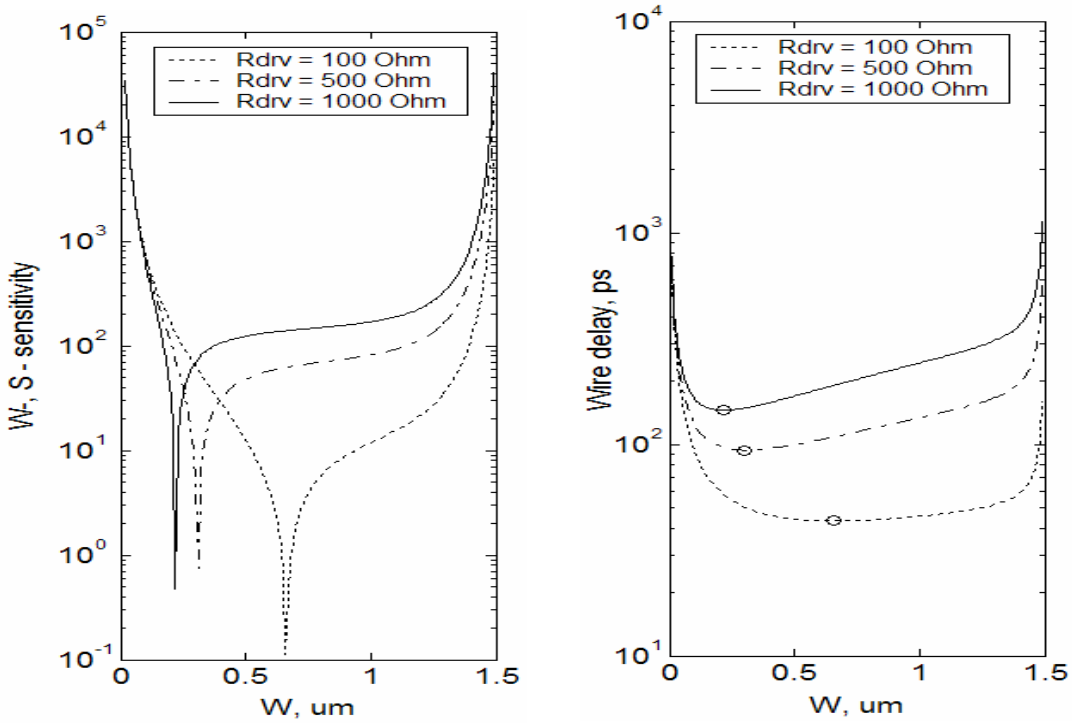


Figure 3: Delay sensitivity to width and space for a single wire

Timing Objectives for bus optimization

We are seeking wire width and space allocation yielding "optimal timing". The definition of optimality depends on the design scenario. Typically, in the early stages of migration there are no firm timing specifications for internal nodes of the circuits. Therefore, the initial goal is to reduce all signal delays. At a later stage, realistic time budgeting is performed such that each signal is assigned a required arrival time. The slack of each signal is defined as the difference between the required arrival time and the delay. Negative slack indicates a violation of specifications. At the final stage, the design is tuned in order to eliminate negative slacks and reduce the delay of the most critical paths. In the following we'll define four commonly used timing objectives.

First objective aims at maximizing the total sum of slacks (same as maximizing the average slack). Let T_i be the required time of the signal σ_i. The objective is thus defined as follows:

$$(10)\ f_1(\overline{W},\overline{S}) = \sum_{i=1}^{n} T_i - \Delta_i(\overline{W},\overline{S}) = \sum_{i=1}^{n}\left[T_i - \left(a_i W_i + b_i + \frac{c_i}{W_i} + \left(d_i + \frac{e}{W_i}\right)\times\left(\frac{1}{S_{i-1}} + \frac{1}{S_i}\right)\right)\right].$$

Total sum of slacks is used when required arrival times are specified. It indicates the potential for increasing the operating frequency of a chip (by reducing the required times T_i). Separation to sum of negative slacks and sum of positive slacks is useful: In migration of old designs into new manufacturing process, the total negative slack reflects the amount of circuit design effort required to meet timing goal. The sum of positive slacks indicates opportunities to save power by downsizing of drivers. This can be done either by reducing transistor width, which reduces dynamic power, or by increasing gate length, thus reducing leakage power.

When required times are still undetermined, an objective of minimizing total sum of delays is commonly used. Notice that from a mathematical point of view this is equivalent to maximizing the first objective, since

$$(11)\ f_2(\overline{W},\overline{S}) = \sum_{i=1}^{n} \Delta_i(\overline{W},\overline{S}) = -f_1(\overline{W},\overline{S}) + \sum_{i=1}^{n} T_i\ .$$

The term $\sum_{i=1}^{n} T_i$ however is constant and doesn't affect the optimization. In the sequel we'll discuss the minimization of total sum of delays f_2. Without loss of generality the results are applicable to maximization of total slack f_1.

Both (10) and (11) are cumulative metrics, integrating the contribution of all signal wires. These are useful objectives for design migration, where the goal is to deliver overall timing speedup. The important factor in such a design scenario is the average speedup, which is well reflected by (10) and (11).

When tuning of critical signals is of interest, other goals are more suitable. This design scenario calls for *MinMax* optimization problems. The objective function is the maximal worst slack or delay among all signals and the optimization attempts to minimize these maxima. Hence, a third objective is to minimize the worst slack among all signals, expressed by f_3 below. Note that we exchanged the terms of the slack for the sake of mathematical convenience.

$$(12)\quad f_3(\overline{W}, \overline{S}) = \max_{1 \le i \le n} \{\Delta_i(\overline{W}, \overline{S}) - T_i\}.$$

A fourth objective aims at minimizing the delay of the slowest signal in the bus. It can be used when timing constraints are not known yet. The corresponding objective function is:

$$(13)\quad f_4(\overline{W}, \overline{S}) = \max_{1 \le i \le n} \{\Delta_i(\overline{W}, \overline{S})\}.$$

In the following we'll explore the optimization of the objective functions f_1 through f_4 by varying the widths and spaces of the bus wires. We first note that all the objectives have a global optimum since the underlying problems are all convex or concave. The convexity proof is given in Appendix A. Additional useful properties of the underlying optimization that suggest

efficient solutions are discussed below. Let us ignore design rules (2) and (3) for the sake of easing the analysis. These do not change the nature of the problem. If the solution of minimizing any of the objectives discussed above subject to (1) satisfies (2) and (3), we are done. If not, search techniques are in order. The convexity of the problem however ensures that no matter what technique is used to solve it, the maximum obtained is optimal.

Optimizing total sum of slacks or delays

We are aiming at minimizing (11) subject to (1). In order to find the minimum of f_2 under g constraint, let us calculate partial derivatives.

$$(14)\ \frac{\partial f_2}{\partial W_i} = a_i - \frac{c_i}{W_i^2} - \frac{e}{W_i^2}\left(\frac{1}{S_{i-1}} + \frac{1}{S_i}\right), i = 1,..., n$$

$$(15)\ \frac{\partial f_2}{\partial S_i} = -\frac{1}{S_i^2}\left[d_i + d_{i+1} + e\left(\frac{1}{W_{i+1}} + \frac{1}{W_i}\right)\right], i = 0,..., n$$

$$(16)\ \frac{\partial g}{\partial W_i} = 1, 1 \le i \le n \ , (17)\ \frac{\partial g}{\partial S_i} = 1, 0 \le i \le n$$

At minimum there exists some real number λ (Lagrange multiplier), satisfying $\nabla f_1 = \lambda \nabla g$. Rearranging and substituting yields the following:

$$(18)\ \lambda = a_i - \frac{1}{W_i^2}\left[c_i + e\left(\frac{1}{S_{i-1}} + \frac{1}{S_i}\right)\right], i = 1,..., n$$

$$(19)\ \lambda = -\frac{1}{S_i^{\,2}}\left[d_i + d_{i+1} + e\left(\frac{1}{W_{i+1}} + \frac{1}{W_i} \right) \right], i = 0,..., n$$

We define $W_0 = W_{n+1} = \infty$ to represent a sidewall connected to power or ground. The above equations plus the area constraint equation (1) impose *2n+2* algebraic equations in *2n+2* variables $\lambda, W_1 \cdots W_n, S_0 \cdots S_n$.

The equations to obtain the maximum of total sum of slacks are identical to (18) and (19). Similar arguments hold, except that minimum is replaced by maximum and convexity by concavity.

Minimizing maximal delays and negative slack: MinMax problems

Objective functions (12) and (13) dealing with worst delay and slack are not differentiable. Therefore, the respective MinMax optimization problems cannot be solved analytically. Although general convex programming or Lagrange relaxation [7] can be employed, we propose a solution approach based on the following properties of these specific problems, yielding an efficient iterative solution with guaranteed convergence.

Theorem 1 *(necessary condition):* In the optimal solution of minimizing the maximal delay in (13) (worst slack in (12)) subject to the area constraint (1), all the delays (slacks) are equal.

Proof: Let us prove the case of delays. Assume on the contrary that the above assertion doesn't hold. Namely, in the optimal solution there exists a wire i whose associated delay is greater than all others. If there are few maximal

ones, pick one having a neighbor with a smaller delay. Such one must exist, as otherwise the delays satisfy the statement of the theorem.

There exist therefore signals σ_{i-1}, σ_i and σ_{i+1}, such that their corresponding delays Δ_{i-1}, Δ_i and Δ_{i+1}, respectively, satisfy $\Delta_{i-1} < \Delta_i$ and $\Delta_{i+1} \le \Delta_i$. We may now narrow wire $i-1$ slightly, thus increasing its delay, say by a magnitude that doesn't exceed $(\Delta_i - \Delta_{i-1})/2$. We may similarly narrow wire $i+1$ and increase its delay by $(\Delta_i - \Delta_{i+1})/2$ if $\Delta_{i+1} < \Delta_i$ indeed. Such narrowing must reduce Δ_i since the width of wire i didn't change, but its spacing from neighbors was increased. Δ_i Which was a maximal delay was thus reduced. If this was the single maximal delay, a contradiction follows since the maximal delay was reduced, while other delays do not exceed it. If there are several wires with maximal delay, the same procedure repeats itself for the next maximal delay wire, until all maximal delays are reduced. This procedure must terminate since the problem it finite.

The proof for objective of worst negative slack follows similarly. •

Theorem 1 imposes necessary conditions on optimal solutions. It is not true that any solution whose delays (or slacks) are all equal is optimal. The convexity of the max objective functions ensures a unique and global minimum. These functions are continuous but not differentiable, so we cannot rely on equating first derivatives to zero in order to express sufficient conditions for optimality. We'll instead attempt to change one of the space or width variables. A single variable however cannot change alone due to the

area constraint. We'll therefore attempt to make a local change of a triplet (S_{i-1}, W_i, S_i) or (W_i, S_i, W_{i+1}), without changing any other variable, such that $S_{i-1} + W_i + S_i$ or $W_i + S_i + W_{i+1}$ are invariant. We define this as an *area preserving local modification*. Clearly, it affects only the delays of $(\sigma_{i-1}, \sigma_i, \sigma_{i+1})$ or (σ_i, σ_{i+1}), respectively. All other delays are unaffected.

Let $\varepsilon > 0$ be arbitrarily small and $0 \le \alpha \le 1$ be real positive numbers. Area preserving local modification of (S_{i-1}, W_i, S_i) will result in the triplet $(S_{i-1} \mp \alpha\varepsilon, W_i \pm \varepsilon, S_i \mp (1-\alpha)\varepsilon)$, for which wire width is increased (decreased), while its neighbor spaces are decreased (increased). Similarly, the modification of (W_i, S_i, W_{i+1}) will result in the triplet $(W_i \mp \alpha\varepsilon, S_i \pm \varepsilon, W_{i+1} \mp (1-\alpha)\varepsilon)$. Notice the correspondence between the plus and minus signs in the modified triplets.

Since max delay (or worst slack) is a convex objective whose global minimum is the MinMax point, the following statement is in order.

Postulate: For any equal delay (or slack) solution other than the MinMax one, there exists an area preserving local modification which reduces the delay (or slack) of a signal without increasing the delay of any other signal.

The following theorem provides a sufficient condition for an equal delay (or slack) solution to be the global minimum.

Theorem 2 *(sufficient condition)**:*** Let all the delays in max delay (worst slack) objective function be equal to each other. This is then the MinMax solution if for all i and any $0 \le \alpha \le 1$ the following relations exist:

$$(20)\; a_i - \frac{c_i}{W_i^2} + \left(d_i + \frac{e}{W_i} \right) \left(\frac{\alpha}{S_{i-1}^2} + \frac{1-\alpha}{S_i^2} \right) - \frac{e}{W_i^2} \left(\frac{1}{S_{i-1}} + \frac{1}{S_i} \right) = 0,$$

$$(21) \left[-a_i + \frac{c_i}{W_i^2} + \frac{e}{W_i^2} \left(\frac{1}{S_{i-1}} + \frac{1}{S_i} \right) \right] - \left(d_i + \frac{e}{W_i} \right) \frac{1}{S_i^2} = 0,$$

$$(22) \left[-a_{i+1} + \frac{c_{i+1}}{W_{i+1}^2} + \frac{e}{W_{i+1}^2} \left(\frac{1}{S_i} + \frac{1}{S_{i+1}} \right) \right] - \left(d_{i+1} + \frac{e}{W_{i+1}} \right) \frac{1}{S_i^2} = 0,$$

where, a_i, c_i, d_i, e are the coefficient of delay equation (5). The proof can be found in Appendix C. Notice that the terms comprising the conditions (20) – (22) are reminiscent of the derivatives in (14) and (15). Hence, an equal delay (slack) solution is optimal if no area-preserving local modification can be found to improve any of the bus wires.

Iterative algorithm for MinMax delay or slack

Theorems 1 and 2, and the convexity properties discussed earlier suggest an iterative algorithm to obtain a minimum of maximal delay (It can be easily adapted to maximize the most critical slack). The algorithm works in two phases which repeat themselves until convergence.

The first phase equates the delay of all signals by iterations. It picks the signal whose delay is currently maximal. It then reduces the delay by equating it with

its two neighbors, a technique used in the proof of Theorem 1. This is repeated until all delays are equal.

The second phase checks for existence of the sufficient condition posted in Theorem 2. It then picks the triplet which mostly violates the sufficient condition and performs an optimal area preserving local modification which is reducing the delay of the triplet's signals.

This gives a rise for another iteration of first phase, as the delay of all the signals can equate at a lower value. If the sufficient condition is satisfied however, the algorithm terminates at optimum.

The algorithm for maximal delay minimization is outlined below. Some heuristics aiming at speeding up convergence are included.

```
MinMaxDelay ( )
set initial solution;
do {
  while ( not all signal delays are equal ) { // first phase
     1. Pick signal with maximal delay;
     2. Equate delay of the selected signal with its
           two neighbors;
  }

  if (sufficient condition fulfilled )
    terminate; // optimum reached
  else // second phase
```

1. Find the triplet which violates the sufficient condition most strongly;

2. Reduce delay of triplet's signal by area preserving local modification;

}

Convergence of the above algorithm can be proven as follows: The inner loop of ***while*** (first phase) iterates over signals and reduces the maximal delay. Therefore, the maximal delay, which is positive, is monotonically decreasing. Hence it must reach a limit. In the outer ***do*** loop the delay (equal for all signals) is also monotonically decreasing, thus it must reach a limit as well.

The relation between minimal total sum and MinMax solutions

We further study the relation between the optimal solutions of total sum and MinMax optimizations, for either delay or slack optimizations. We may interpret the delay (slack) of the bus $\Delta \equiv \left(\Delta_1,...,\Delta_n\right)$ (analogously for slacks) as a vector in n dimensional vector space over real positive numbers. The addition of delay (slack) vectors is interpreted as connecting two busses serially, signal by signal. It is not difficult to prove that the objective function of total sum of slacks (10) or delays (11), and the objective function of max slack (12) or delay (13) are nothing but the norms $\| \|_1$ and $\| \|_\infty$, respectively. Let $v \in V$ be any vector in n dimensional vector space V. The *norm equivalence theorem* states that there exist real positive numbers α and β satisfying $\alpha < \|v\|_1 / \|v\|_\infty < \beta$. This means that an optimal solution of minimizing the total sum of delays is also a good MinMax solution and vice versa. Indeed, the following theorems claim that the optimal solution of the MinMax problem is bounded from both sides by the optimal solution of the total sum problem. The notation is shown in Figure 4, illustrating distributions

of signal delays in the solution of a minimal total delay problem and in the solution of the corresponding MinMax delay problem.

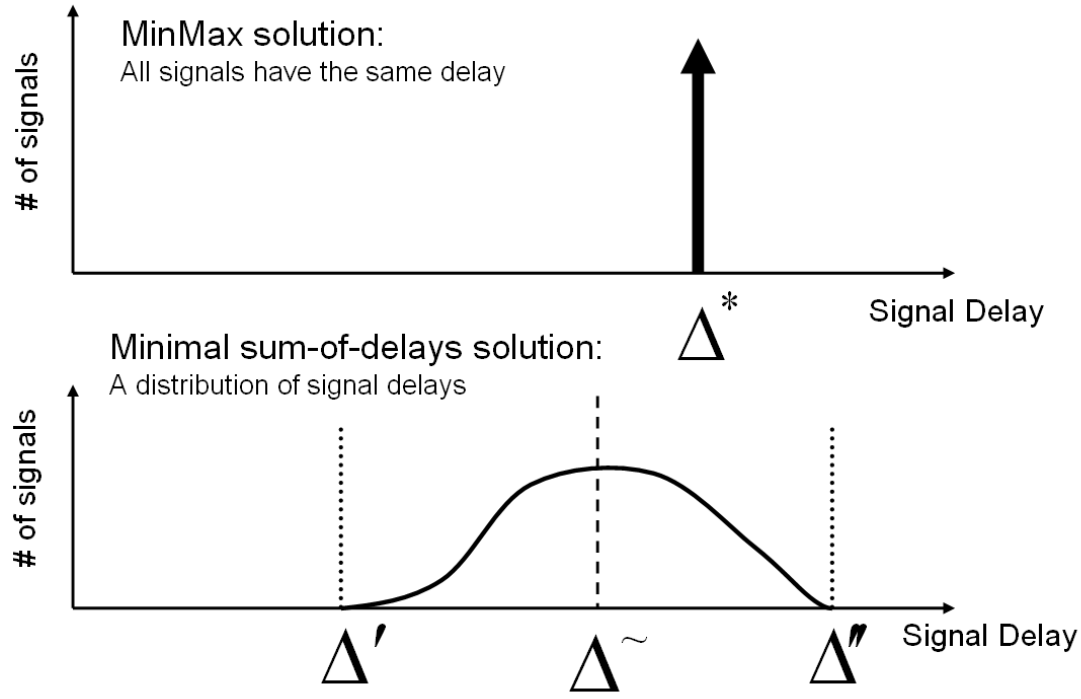


Figure 4: Distributions of signal delays in MinMax solution (top) compared with minimal sum-of-delays solution (bottom).

Theorem 3: Let Δ', $\Delta^{\sim}$ and Δ'' be the smallest, average and largest delay, respectively, among all the bus signals in the optimal solution of minimal total sum of delay. Let Δ^* be the delay of each signal in the MinMax optimal solution. There exists then $\Delta' \le \Delta^{\sim} \le \Delta^* \le \Delta''$.

Proof: The inequality $\Delta' \le \Delta^{\sim} \le \Delta''$ is satisfied by definition. It is impossible that $\Delta^* < \Delta^{\sim}$. Otherwise, the optimal MinMax solution yields total sum of delay $n\Delta^*$, thus contradicting the optimality of $n\Delta^{\sim}$. It is also impossible that

$\Delta'' < \Delta^*$ as it yields a solution whose max delay is smaller than Δ^*, contradicting the optimality of Δ^*. ●

Theorem 4: Let Ω', $\Omega\ \tilde{}$ and Ω'' be the smallest, average and largest slack of a signal, respectively, in the optimal solution of maximal total sum of slack. Let Ω^* be the slack of a signal in the MinMax optimal solution. There exists then $\Omega' \le \Omega\tilde{} \le \Omega^* \le \Omega''$.

Examples

Exampel 1: Sidewall effects in a uniform bus.

Figures 5 and 6 illustrate the optimal solutions of MinMax and sum-of-delays optimization, respectively. The bus has eight signals whose wire length is 500 microns. All drivers are of 500Ω resistance and all load capacitances are 50 fF. The area allocated for the bus is 7 microns.

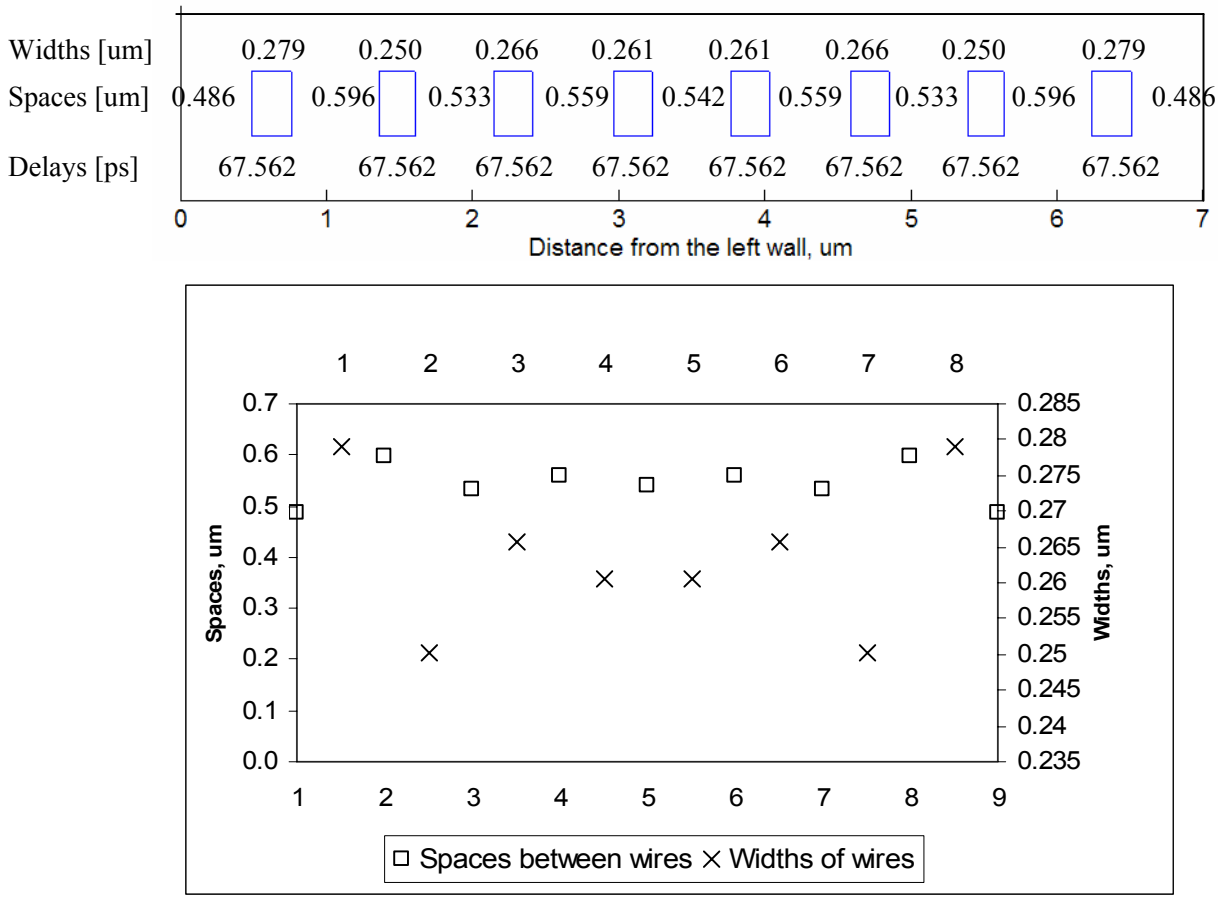


Figure 5: (Top): Cross section of the bus after MinMax delay optimization, annotated with values of wire widths and spaces. (Bottom): Width and spaces shown as graphs versus wire position in the bus.

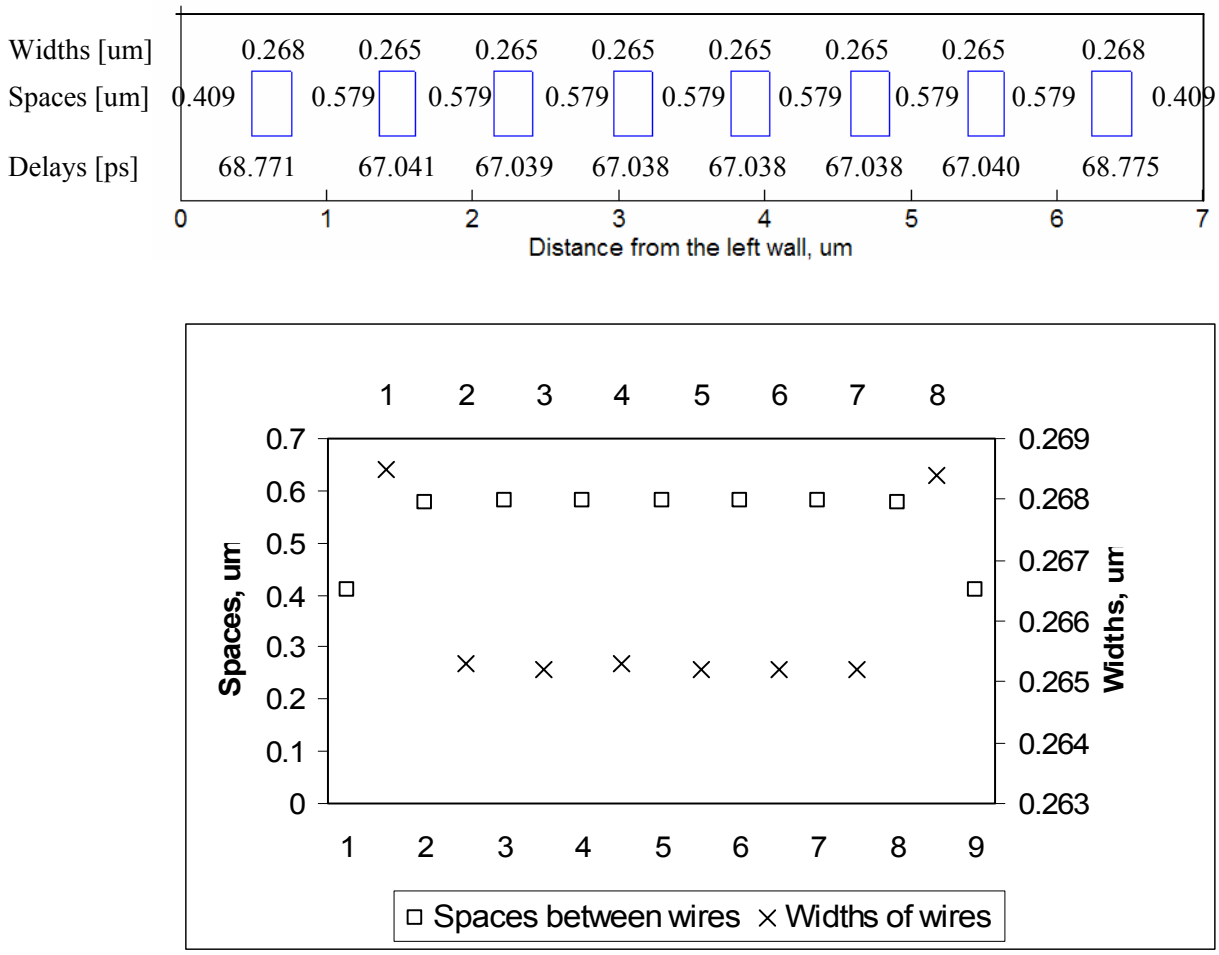


Figure 6: (Top): Cross section of the bus after sum-of-delays optimization, annotated with values of wire widths and spaces. (Bottom): Width and spaces shown as graphs versus wire position in the bus.

In case of MinMax optimization all signal delays are identical as expected. Notice that wire width and space have 'oscillations' decaying towards the center of the bus. This is caused by the side walls, which get relatively small spaces to the extreme wires, because unlike all other spaces their cross-capacitance is not shared by two signals. The narrow space needs

compensation by a wide wire, otherwise large RC delay will occur. This phenomenon repeats itself for the next adjacent wires, with decreasing amplitude. In the minimization of sum-of-delays, the first and last wires are affected similarly as in MinMax optimization due to same reason: sidewalls don't care for space. All other signals however have the same width and space. Consequently, the extreme wires have larger delay than all others. Despite differences in width-space distributions between two cases, numerical values of delays are very close. Comparing average delay obtained in sum-of-delays optimization with delay obtained by MinMax optimization yield $\Delta^{\sim} = 67.473 psec$ and $\Delta^{*} = 67.562 psec$, which are indeed very close.

For deeper insight while comparing total sum-of-delays with MinMax problems, let's simplify the bus model and ignore the sidewall effect. This is done by dropping the sidewalls and assuming that the leftmost signal and the rightmost signal are adjacent. Pictorially, it is equivalent to placing the signal bus on a cylindrical surface, thus obtaining two neighbors for every signal. The optimal solution satisfies the following theorem whose proof is given in Appendix B.

Theorem 5: Let all signals have identical drivers and identical receivers and let their order be cyclical (placed on a cylindrical surface). Then in the optimal solution of maximizing (minimizing) the total sum of slacks (delays), all the widths, spaces and delays are necessarily equal.

We now characterize the optimal solution of MinMax delay in a cyclical uniform bus by a direct consequence of Theorems 3 and 5 above.

Corollary 1: For a cyclical bus where all signals have identical drivers and identical receivers, the minimization of max delay yields the same solution as the minimization of total sum of delays.

Proof: Follows directly from Theorem 3 which states that $\Delta' \le \Delta^{\sim} \le \Delta^{*} \le \Delta''$, where Δ', $\Delta^{\sim}$ and Δ'' are the smallest, average and largest delays in the minimal total sum of delays, respectively, and Δ^{*} is the delay of a signal in the optimal MinMax solution. Theorem 5 states that for cyclic uniform bus there exists $\Delta' = \Delta^{\sim} = \Delta''$. Hence the corollary follows.●

Returning to Example 1 above, let us modify the bus to be cyclical. Both MinMax optimization and minimal sum-of-delay optimization were solved in MATLAB and yielded a result $65.524\,psec$. In conclusion, a uniform bus is similar to a cyclical bus, except for the edge effects near the sidewalls. Therefore, optimal solutions for total delay and MinMax delay are almost identical. Note that the identity of optimal solutions for total sum and MinMax doesn't exist for slacks, even in a uniform cyclical bus. Maximizing total sum of slacks is the same as minimizing total sum of delays; hence delays of signals are all equal in the optimized uniform cyclical bus. Slacks, however, are not equal to each other as they depend on the required time which may change from signal to signal. In the optimal solution of the MinMax slack problem, all slacks are equal.

The next example deals with optimizing total slack and worst slack in a uniform bus with side walls.

Example 2: Slack optimization.

Figures 7 illustrates the case where a required time is assigned to each signal. Using the same bus from example 1, a required time of 65 picoseconds was assigned to the fifth wire, while all other wires were allowed 70 picoseconds. Applying MinMax optimization of the slacks results in equal slacks of 1.4 picoseconds for all signals, as shown in Figure 7. The distribution of wire widths and spaces is depicted in the bottom part of Figure 7. Its nature is similar to the case of MinMax delay optimization. Notice however that the non-uniformity in required time disturbs the symmetry obtained in Example 1. The wire which was assigned the most difficult (earliest) required time became wide, while its spacing to adjacent wires became larger too. This is for the sake of reducing its RC delay, thus compensating for the early required time.

Recall that maximizing total sum of slack is not affected by required times as this optimization is identical to minimizing total delay, as discussed earlier.

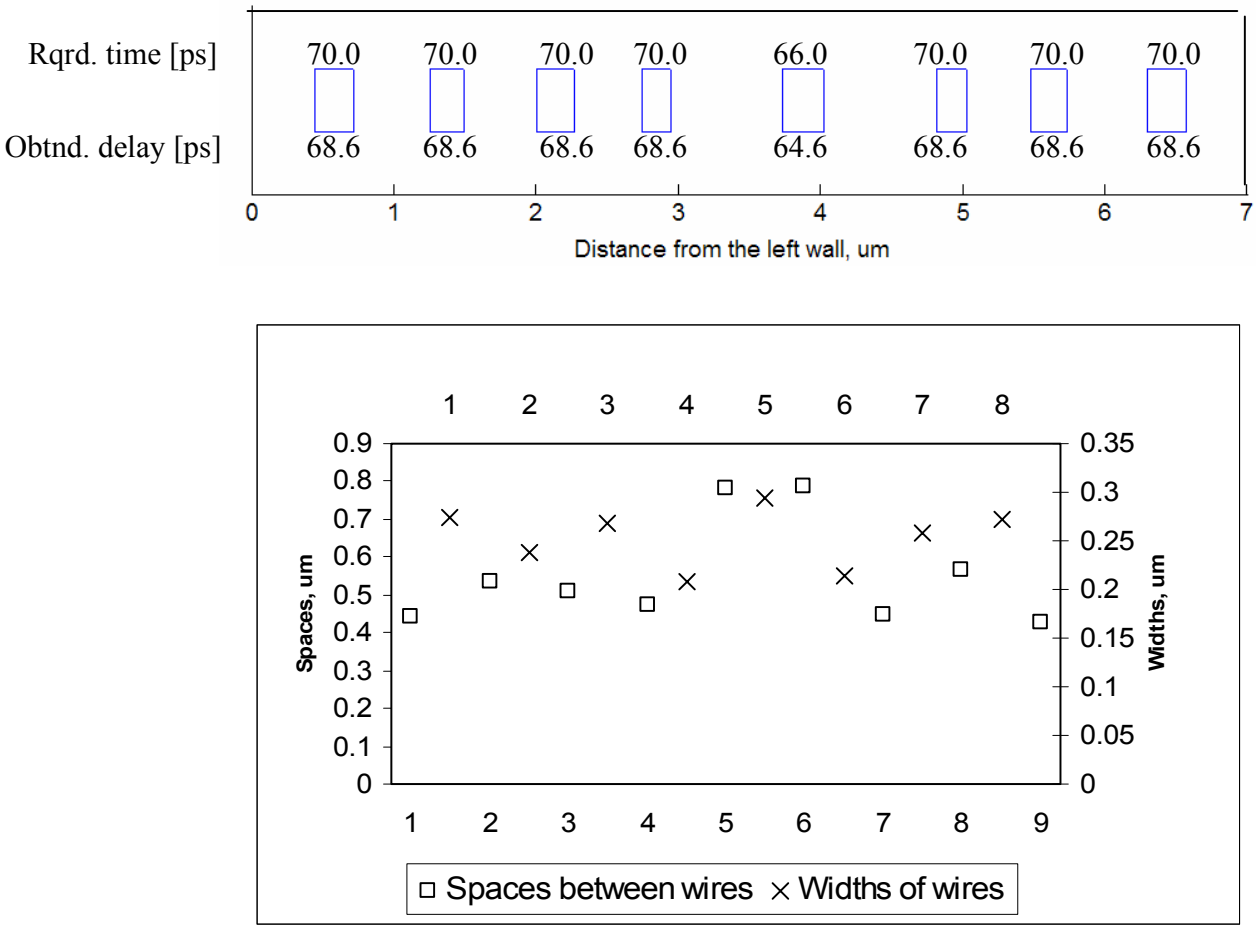


Figure 7: (Top): Cross section of the bus after MinMax slack optimization. (Bottom): Width and spaces shown as graphs versus wire position in the bus.

Example 3: Non-uniform bus

This example presents an interleaved bus structure, with alternating drivers; odd-numbered wires have strong drivers (R_d=100Ω), and even-numbered wires have weak drivers (R_d=1000Ω). Wire length is 3000μm and load capacitance 70 fF in this example. A total bus width is constrained to 10 microns. Results of optimizing this bus are depicted in Figures 8 and 9. Notice first that the wire width and space distributions of MinMax optimization differ

significantly from total sum-of-delays minimization. Since in MinMax optimization all the delays must be equal, and since the weak and strong drivers are interleaved, the spaces must be equal to each other. An exception is the leftmost space. This is due to the asymmetry resulting from a strong driver on the left side and a weak driver on the right side of the bus. The equality of spaces and signal delays implies that signals with strong drivers will be narrower than those driven by weak drivers, as demonstrated in the bus cross section. Minimization of total sum-of-delay also yields alternating widths of wires, but neither uniformity nor symmetry exists. Notice also that wider wires were allocated to strong drivers in this case. It is interesting to compare the delays obtained by the two optimizations. Although all the relations proved in Theorem 3 do exist, the MinMax delay is much worse than the average delay in the total sum-of-delay optimization. In fact, it is very close to the maximal delay of the latter distribution.

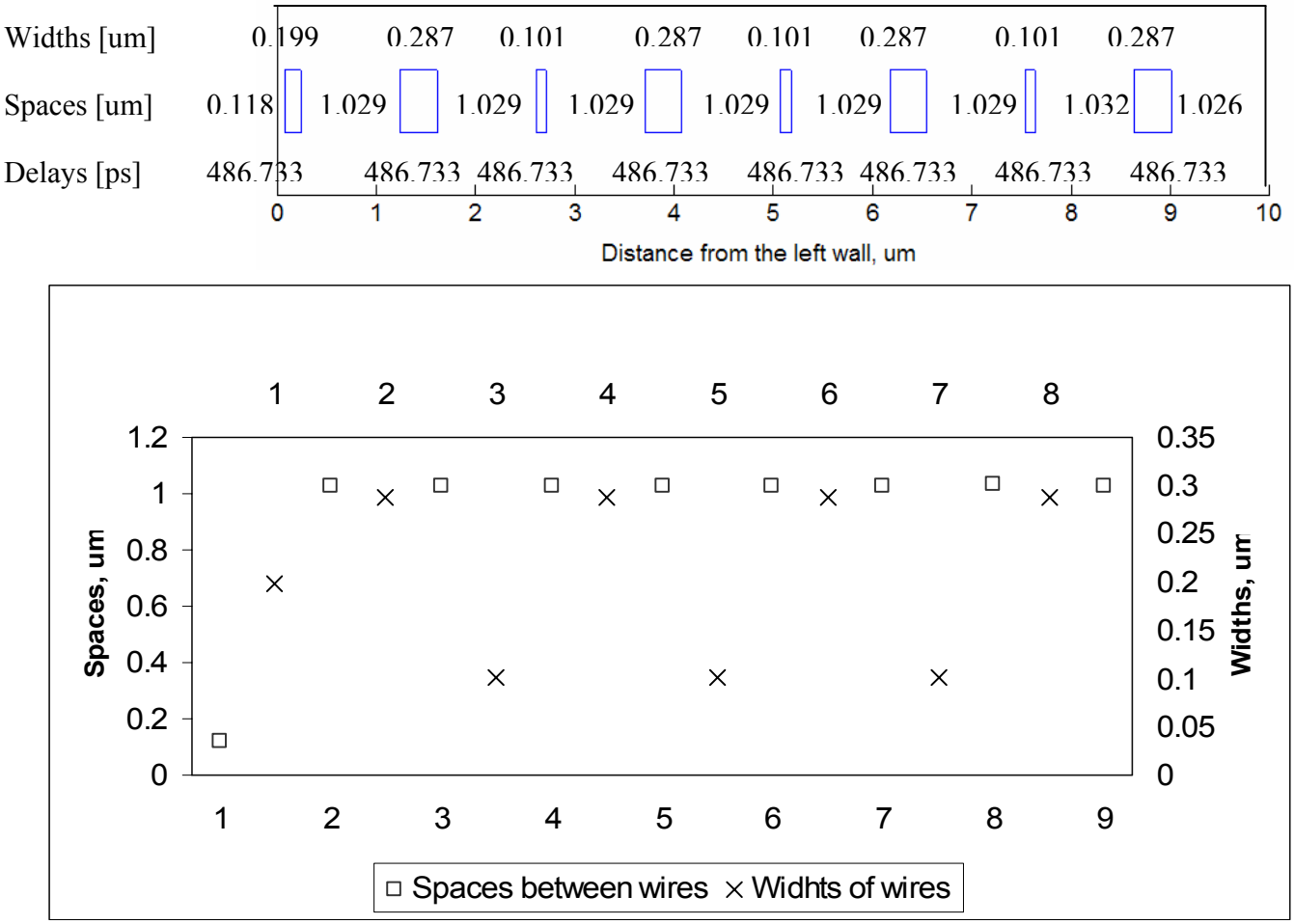


Figure 8: Non uniform bus minmax optimization

Example 5: Optimal delay dependency on bus width.

Let us change the total width of the bus in example 4. The MinMax delay is compared to the minimum, average and maximum delay of the corresponding total sum-of-delays optimization, for various bus widths, as illustrated in Figure 9. According to Theorem 3, the MinMax delay of all wires always resides between the average and the maximal wire delay in the total sum-of-delay minimization. As the bus width constraint is relaxed (larger widths), the MinMax result approaches the maximal delay of the other optimization. This

is due to the fact that large bus width decouples the signals, so signals of weak and strong drivers are optimized independently.

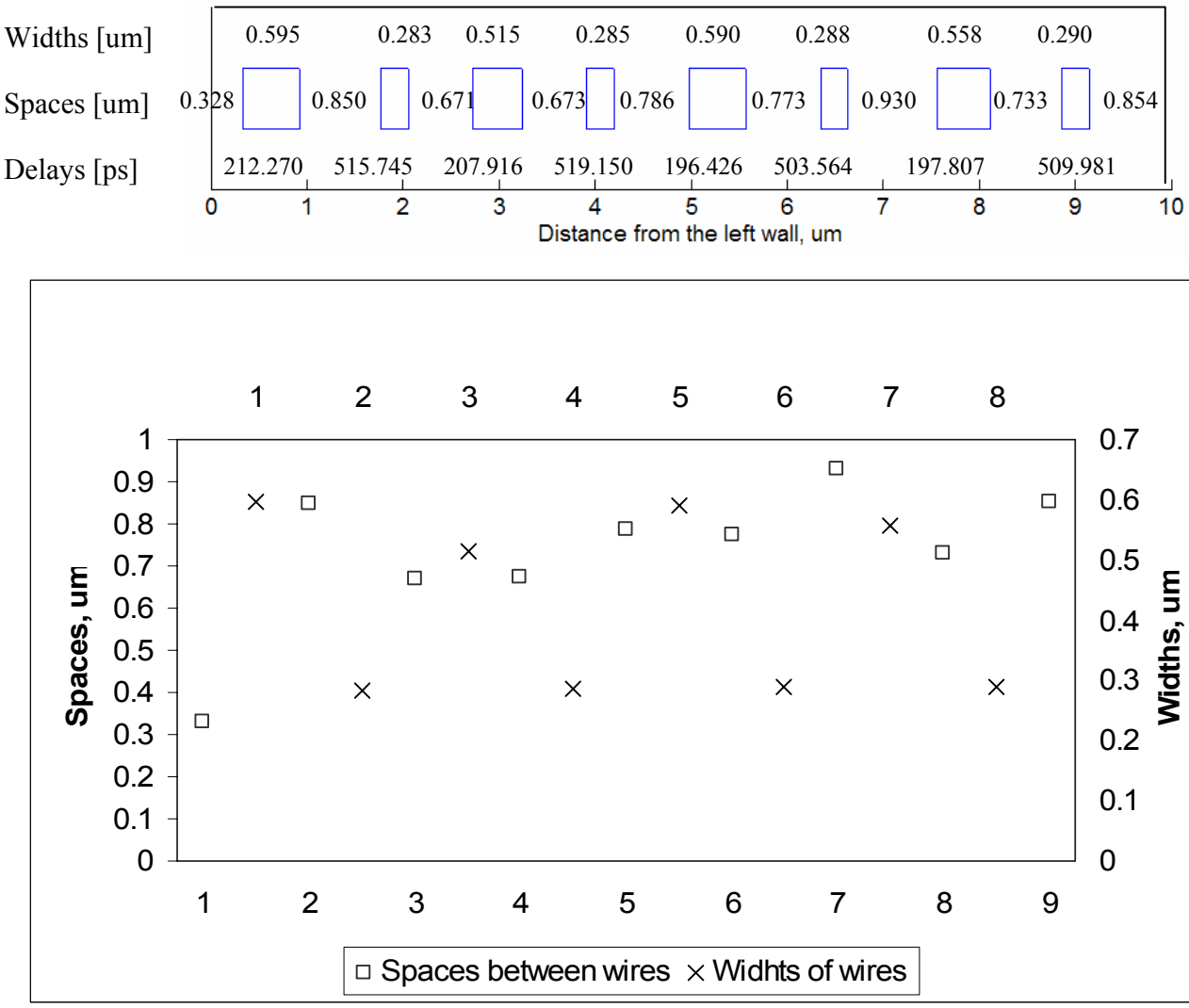


Figure 9: Non uniform bus total sum of delays optimization

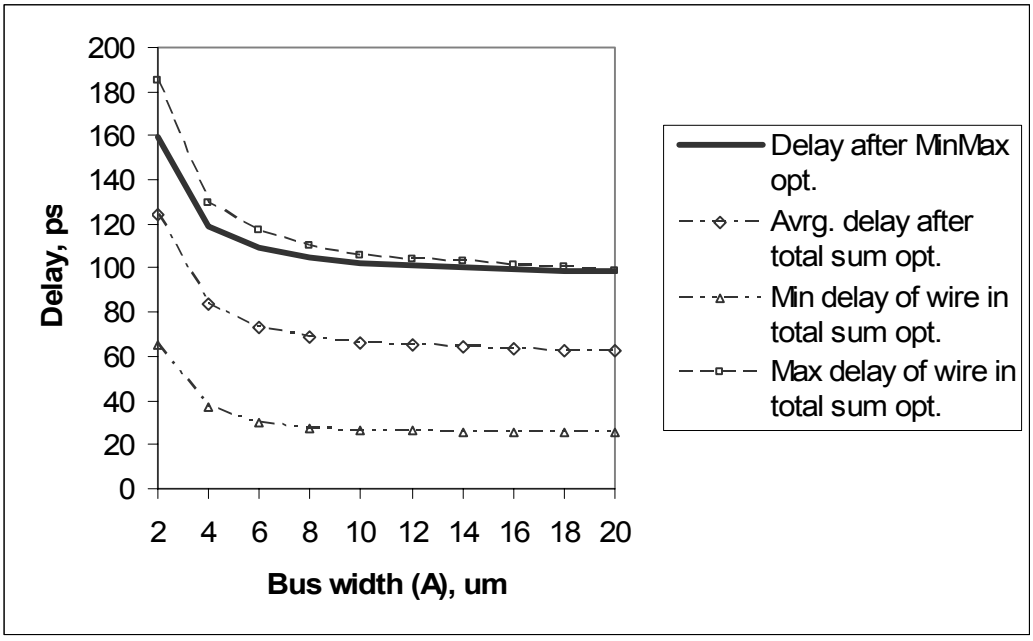


Figure 10: Optimal solution parameters Δ', $\Delta^{\sim}$, Δ^{*}, Δ'' (see Figure 4) versus bus width constraint A, for the circuit of example 4.

Example 6. Migration of a bus in an industrial circuit.

A 20-wire bus structure from an industrial circuit block was migrated from 130 nanometer to 90 nanometer technology. The total width of the bus is 10.615 μm, length of wires is 200 μm. Before optimization, wire widths and spaces were determined by shrinking the old layout, as specified in Table I below.

Table I: Migration of a circuit block in 90 nanometer technology

Wire no.(i)	R_{driver} [$K\Omega$]	C_{load} [fF]	Before optimization			After optimization		
			S_{i-1} [um]	W_i [um]	Delay [ps]	S_{i-1} [um]	W_i [um]	Delay [ps]
1	0.110	22.0	0.163	0.330	16.589	0.162	0.239	16.539
2	0.123	18.9	0.225	0.330	16.747	0.223	0.205	16.627
3	0.042	114.0	0.110	0.330	23.453	0.180	0.468	20.593
4	0.116	14.5	0.225	0.330	15.552	0.197	0.217	15.321
5	0.038	131.4	0.110	0.330	24.794	0.185	0.543	20.619
6	0.122	18.6	0.225	0.330	15.691	0.201	0.212	16.586
7	0.040	120.7	0.185	0.330	23.459	0.181	0.544	20.115
8	0.098	21.1	0.225	0.330	14.810	0.182	0.226	15.748
9	0.104	14.4	0.225	0.330	13.905	0.227	0.204	14.491
10	0.117	15.4	0.225	0.330	14.891	0.232	0.212	15.292
11	0.045	110.1	0.180	0.330	23.059	0.200	0.505	20.414
12	0.039	112.9	0.195	0.330	22.444	0.143	0.474	20.165
13	0.125	16.9	0.210	0.330	15.324	0.185	0.211	15.976
14	0.122	16.1	0.225	0.330	16.150	0.254	0.209	15.624
15	0.042	111.2	0.110	0.330	23.132	0.192	0.464	20.397
16	0.122	17.2	0.225	0.330	15.171	0.180	0.198	16.386
17	0.049	111.1	0.225	0.330	23.487	0.207	0.499	20.955
18	0.122	14.3	0.225	0.330	14.660	0.181	0.196	15.737
19	0.040	121.6	0.215	0.330	23.467	0.200	0.529	20.241
20	0.110	15.3	0.225	0.330	15.364	0.195	0.198	15.812
Total					**372.149**			**353.639**

Total sum of delays timing optimization was run on this bus and results of wire resizing and timing analysis are presented in Table I. Minimization of maximum delay also has been performed, yielding a result of 18.942 ps

(compare to the average wire delay 17.685 ps and maximum wire delay 20.619 ps in sum of delays optimization).

As seen in the table, average timing of the bus was improved by about 5%. It was achieved by decreasing widths of wires, thus decreasing wire capacitances. However, some wires with strong drivers were widened to decrease their resistance. Since these wires drive large capacitances, their resistance is important.

Discussion and conclusions

It has been noted in general circuit timing optimization that whenever MinMax formulation is used, all timing paths tend to become equally critical, as proven in theorem 1 for the bus sizing problem. This phenomenon was termed path-balancing [16][17]. Shaping of delay distributions involves a number of considerations: minimization of critical delay for maximal circuit speed tends to push the right tail of the path delay distribution towards the center, and minimization of area and power often tends to push the left tail towards the center. Circuit tuning by MinMax optimization results in a very narrow distribution of delays, as illustrated in figure 4. However, a narrow distribution where many paths are critical, is more sensitive to parameter variability [16,17,18]. A modification of the objective function in MinMax formulation has been proposed, involving a penalty function [17] which separates the most critical paths from others. An alternative approach is to use the total sum objective, at least in the initial stages of layout migration. At the optimal the solution of total sum minimization, every wire is at a balance point where sensitivity to the value of wire width and spaces approaches zero, as illustrated in Figure 3. In contrast, the MinMax solution takes most signals away from this stable point in order to equalize all of the delays or slacks, and

therefore the circuit becomes more sensitive to variations in geometrical dimensions. Our computational experiments in bus tuning show that typically, most signal delays become much worse while the critical signal become only slightly better as resources are shifted to make all wires equally critical. In other words, the largest wire delay in the total sum solution is typically a tight bound on the solution of the MinMax problem, as seen in the example of Figure 9.

In conclusion, we have characterized the problem of simultaneously allocating wire widths and spaces to all wires in parallel bus structures under a total area constraint, for circuit performance optimization. We have demonstrated its importance in migration of layouts to new generations of CMOS process technology. Our results show that total sum of delays (or slacks) is a useful objective function for minimization. Compared with MinMax delay tuning, it is mathematically more convenient, leads to robust solutions which are less sensitive to parameter variations, and typically produces bus layouts which closely approach the best achievable performance. We have also presented an iterative algorithm for MinMax performance tuning of bus layouts, based on problem-specific necessary and sufficient conditions for optimality.

Appendix A: The optimization problems are all convex

Proposition: The objectives functions f_1, f_2, f_3, f_4 presented above are all convex or concave. Their associated constraints are linear, and therefore also convex. Consequently all the optimization problems have unique, global minimum or maximum, depending on convexity or concavity.

Proof: The area and design rules constraints given in (3), (4) and (5) are all linear equalities or inequalities. Altogether they define a convex feasible region on which the above objective functions are defined.

The function $\Delta_i\left(\overline{W}, \overline{S}\right)$ in (2) is a sum of terms depending on the variables W_i, S_i and S_{i-1}. In order to prove its convexity it is sufficient to see the convexity for each term, since a linear combination with positive coefficients of convex functions is convex too. Convexity exists if all its second order derivatives are non-negative. Deriving twice all the terms comprising Δ yields $\frac{\partial W}{\partial^2 W} = 0$; $\frac{\partial 1/W}{\partial^2 W} = \frac{2}{W^3} > 0$; $\frac{\partial 1/WS}{\partial^2 W} = \frac{2}{W^3 S} > 0$; $\frac{\partial 1/WS}{\partial W \partial S} = \frac{\partial 1/WS}{\partial S \partial W} = \frac{1}{W^2 S^2} > 0$; $\frac{\partial 1/WS}{\partial^2 S} = \frac{2}{WS^3} > 0$, which are all nonnegative. Consequently, the delay of single signal is convex

The function f_1 defined in (10) is a negative sum of convex terms and therefore yields a concave function. Its maximization on convex region yields

unique global maximum. For similar reasons, f_2 defined in (11) is convex and has therefore global minimum.

Both functions f_3 and f_4 of maximal slack delay as defined in (12) and (13), respectively, are convex since they are a maximum of convex functions. Their minimization on convex region yields therefore a unique global minimum.●

Appendix B: Proof of Theorem 5

Proof: From (11) we obtain:

$$(23)\ f_2 = a\sum_{i=1}^{n} W_i + nb + c\sum_{i=1}^{n}\frac{1}{W_i} + 2d\sum_{i=1}^{n}\frac{1}{S_i} + e\left[\sum_{i=2}^{n}\frac{1}{W_i}\left(\frac{1}{S_{i-1}}+\frac{1}{S_i}\right)+\frac{1}{W_1}\left(\frac{1}{S_n}+\frac{1}{S_1}\right)\right].$$

In the above we identify S_0 with S_n due to the cyclic ordering of signals. Note also that the coefficients are not indexed since all drivers and receivers are identical.

Assume that the optimization problem (3), (18) and (19) was solved and the optimal solution is given. Let $W = \sum_{j=1}^{n} W_j$ and $S = \sum_{j=1}^{n} S_j$ denote the total wire widths and total spacing in the optimal solution. Obviously, there exists $A = W + S$. Let us show that among all the settings of area preserving W and S, the one in which all W_i and S_i are identical is optimal.

Examination of (23) shows that it consists of the following sums $\sum_{i=1}^{n} 1/W_i$, $\sum_{i=1}^{n} 1/S_i$ and $t(\overline{W}, \overline{S}) = \sum_{i=2}^{n}\left[1/W_i\left(1/S_{i-1} + 1/S_i\right)\right] + 1/W_1\left(1/S_n + 1/S_1\right)$. The first

two sums are minimized only when all W_i are equalized and all S_i are equalized. We'll show the term $t(\overline{W}, \overline{S})$ is also minimized by such equalization.

Substitution of $W_n = W - \sum_{j=1}^{n-1} W_j$ and $S_n = S - \sum_{j=1}^{n-1} S_j$ in $t(\overline{W}, \overline{S})$ and then differentiating by each of the $2n-2$ variables $W_1,...,W_{n-1}, S_1,...,S_{n-1}$ yields:

$$(24)\ \frac{\partial t}{\partial W_i} = \frac{1}{W_i^{\,2}}\left(\frac{1}{S_{i-1}} + \frac{1}{S_i}\right) + \frac{1}{\left(W - \sum_{j=1}^{n-1} W_j\right)^2}\left(\frac{1}{S_1} + \frac{1}{S - \sum_{j=1}^{n-1} S_j}\right) = 0$$

$$(25)\ \frac{\partial t}{\partial S_i} = \frac{1}{S_i^{\,2}}\left(\frac{1}{W_i} + \frac{1}{W_{i+1}}\right) + \frac{1}{\left(S - \sum_{j=1}^{n-1} S_j\right)^2}\left(\frac{1}{W_{n-1}} + \frac{1}{W - \sum_{j=1}^{n-1} W_j}\right) = 0 .$$

Note that in both (24) and (25) the second term is identical and independent of i for all equations. Consequently, all the derivatives of $t(\overline{W}, \overline{S})$ by widths satisfy same equation, and so the derivatives by spacing. Therefore, there exist two real numbers λ_w and λ_s, satisfying $1/W_i^{\,2}\left(1/S_{i-1} + 1/S_i\right) = \lambda_w$ and $1/S_i^{\,2}\left(1/W_i + 1/W_{i+1}\right) = \lambda_s$. This implies that $W_1 = W_2 = \ldots = W_n$ and $S_1 = S_2 = \ldots = S_n$ is indeed the optimal and unique solution.

In conclusion, minimal total sum of delays requires identical wire widths and identical wire spacing. Identical signal delays follow immediately due to identical drivers and identical receivers. ●

Appendix C: Proof of Theorem 2

Proof: Assume to the contrary that a given equal delay solution is not minimal. According to the above postulate there exists area preserving local modifications that will reduce the delay of a signal without increasing the delay of any other signal. Four modifications are possible: Increasing or decreasing wire width, and increasing or decreasing a space. Let us consider each.

Case 1: Increasing wire width is impossible since area preservation implies that at least one of the adjacent spaces is decreased. This however increases the delay of the adjacent signal that shares this space.

Case 2: Decreasing the width results in the new triplet $\left(S_{i-1}+\alpha\varepsilon, W_i-\varepsilon, S_i+(1-\alpha)\varepsilon\right)$, where $0\le\alpha\le 1$. The delays Δ_{i-1} and Δ_{i+1} do not increase as their adjacent spaces do decrease. For the new delay Δ_i' to decrease there must exist some $0\le\alpha\le 1$, such that substitution of the new width and spaces in (2) yields

$$0\le\Delta_i-\Delta_i'=\varepsilon\left[a_i-\frac{c_i}{W_i^2}+\left(d_i+\frac{e}{W_i}\right)\left(\frac{\alpha}{S_{i-1}{}^2}+\frac{1-\alpha}{S_i{}^2}\right)-\frac{e}{W_i^2}\left(\frac{1}{S_{i-1}}+\frac{1}{S_i}\right)\right]+O\left(\varepsilon^2\right)$$

Dropping the term $O\left(\varepsilon^2\right)$implies that delay reduction requires the above square brackets to be positive. Case 1 can be viewed as being obtained by using negative ε , thus implying an opposite inequality than the above. Hence equation (20) follows.

Case 3: Decreasing a space results in the new triplet $(W_i + \alpha\varepsilon, S_i - \varepsilon, W_{i+1} + (1-\alpha)\varepsilon)$. We require that none of Δ_i and Δ_{i+1} is increased. This implies the for some $0 \le \alpha \le 1$, there exists

$$0 \le \Delta_i - \Delta_i' = \varepsilon\left\{\alpha\left[-a_i + \frac{c_i}{W_i^{\,2}} + \frac{e}{W_i^{\,2}}\left(\frac{1}{S_{i-1}} + \frac{1}{S_i}\right)\right] - \left(d_i + \frac{e}{W_i}\right)\frac{1}{S_i^{\,2}}\right\} + O(\varepsilon^2)$$

$$0 \le \Delta_{i+1} - \Delta_{i+1}' = \varepsilon\left\{(1-\alpha)\left[-a_{i+1} + \frac{c_i}{W_{i+1}^{\,2}} + \frac{e}{W_{i+1}^{\,2}}\left(\frac{1}{S_i} + \frac{1}{S_{i+1}}\right)\right] - \left(d_{i+1} + \frac{e}{W_{i+1}}\right)\frac{1}{S_i^{\,2}}\right\} + O(\varepsilon^2)$$

Dropping the term $O(\varepsilon^2)$ implies that delay reduction requires the above square brackets to be positive. Hence (21) and (22) must be non negative.

Case 4: Increasing a space results in the new triplet $(W_i - \alpha\varepsilon, S_i + \varepsilon, W_{i+1} - (1-\alpha)\varepsilon)$. This is exactly the same as case 3, but with negative ε. Therefore, the braces need now be non positive. ●

References

1. R. Ho, K. Mai and M. Horowitz, “The future of wires”, *Proceedings of the IEEE*, Vol. 89, no. 4, Apr. 2001.

2. D. Sylvester and K. Keutzer, “Getting to the bottom of deep submicron”, *in Proc. ICCAD*, pp. 203-211, 1998.

3. Kahng, S. Muddu, E. Sarto and R. Sharma, "Interconnect Tuning Strategies for High-Performance ICs", *Proc. Design, Automation and Testing in Europe*, Paris, February 1998.

4. J. Cong and K. S. Leung, "Optimal Wiresizing Under Elmore Delay Model". *IEEE Trans. on Computer-Aided Design*, vol. 14, no. 3, pp. 321-336, Mar. 1995.

5. S. S. Sapatnekar, "Wire Sizing as a Convex Optimization Problem: Exploring the Area Delay Tradeoff*", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, Vol. 15, No. 8, pp. 1001 - 1011, August 1996.

6. C. Chu and D. F. Wong, “Closed Form Solution to Simultaneous Buffer Insertion / Sizing and Wire Sizing”, *ACM Transactions on Design Automation of Electronic Systems*, vol. 6, no. 3, pages 343-371, July 2001.

7. C.P. Chen, C. N. Chu and D. F. Wong, “Fast and exact simultaneous gate and wire sizing by Lagrangian relaxation”, *in Proc. ICCAD* , pp. 617-624, 1998.

8. J. Cong, L. He, C. K. Koh and Z. Pan, "Interconnect Sizing and Spacing with Consideration of Coupling Capacitance", *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems,* vol. 20, no. 9, pp.1164-1169, September 2001.

9. J. Lillis, C. K. Cheng, and T. T. Y. Lin, “Simultaneous routing and buffer insertion for high performance interconnect,” in *Proc. 6th Great Lakes Symp. VLSI*, Mar. 1996, pp. 148–153.

10. M. Becer, D. Blaauw, V. Zolotov, R. Panda, I. Hajj, “Analysis of Noise Avoidance Techniques in Deep-Submicron Interconnects Using a Complete Crosstalk Noise Model”, *IEEE/ACM Design Automation and Test in Europe Conference (DATE)*, March 2002, pp. 456-463.

11. D. A. Kirkpatrick and A. L. Sangiovanni-Vincentelli. “Techniques for crosstalk avoidance in the physical design of high performance digital systems”, *In Proc. IEEE/ACM Int. Conf. Computer Aided Design*, pp. 616--619, Nov. 1994.

12. R. Arunachalam, E. Acar, and S. R. Nassif, “Optimal shielding/spacing metrics for low power design”, *Proceedings of IEEE Computer Society Annual Symposium on VLSI*, pp. 167-172, 2003.

13. J. Cong, D. Z. Pan and P. V. Srinivas, "Improved Crosstalk Modeling for Noise Constrained interconnect Optimization", *ACM/IEEE International Workshop on Timing Issues in the Specification and Synthesis of Digital Systems,* pp. 14-20, Austin, December 2000.

14. A. Abou-Seido, B. Nowak and C. Chu, “ Fitted Elmore Delay: A Simple and Accurate Interconnect Delay Model*”, IEEE Transactions on VLSI Systems*, vol. 12, no. 7, pages 691-696, July 2004.

15. P. Chen, D. A. Kirkpatrick and K. Keutzer, “Miller Factor for Gate-Level Coupling Delay Calculation”, *in Proc. ICCAD*, pp.68-74, 2000.

16. M. Hashimoto and H. Onodera, ”Increase in delay uncertainty by performance optimization”, *Proc. IEEE International Symposium on Circuits and Systems (ISCAS)*, pages 379--382, 2001.

17. X. Bai, C. Visweswariah, P. N. Strenski, D. J. Hathaway, "Uncertainty-Aware Circuit Optimization”, *The 39th Design Automation Conference (DAC)*, 2002, pp.58~63.

18. S. Borkar, T. Karnik, S. Narendra, J. Tschanz, A. Keshavarzi and V. De: ” Parameter variations and impact on circuits and microarchitecture”, *Design Automation Conference (DAC)* 2003, pp. 338-342.